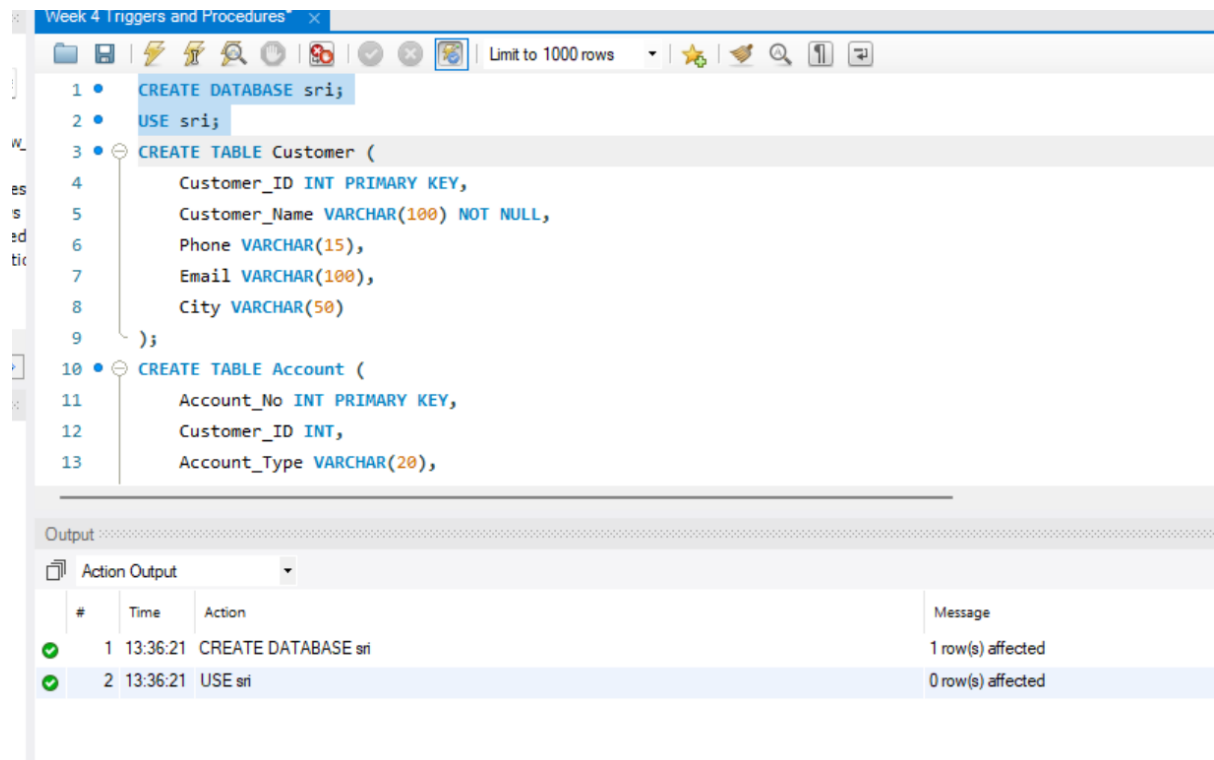


CREATE DATABASE BankDB;

USE Bankdb;



The screenshot shows a SQL IDE window titled "Week 4 Triggers and Procedures". The main editor displays the following SQL code:

```
1 • CREATE DATABASE sri;
2 • USE sri;
3 • CREATE TABLE Customer (
4     Customer_ID INT PRIMARY KEY,
5     Customer_Name VARCHAR(100) NOT NULL,
6     Phone VARCHAR(15),
7     Email VARCHAR(100),
8     City VARCHAR(50)
9 );
10 • CREATE TABLE Account (
11     Account_No INT PRIMARY KEY,
12     Customer_ID INT,
13     Account_Type VARCHAR(20),
```

Below the editor is the "Output" pane, which shows the execution results:

#	Time	Action	Message
✓ 1	13:36:21	CREATE DATABASE sri	1 row(s) affected
✓ 2	13:36:21	USE sri	0 row(s) affected

```
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    Email VARCHAR(100),
    City VARCHAR(50)
);
```

The screenshot shows a database management tool interface. The top toolbar includes icons for file operations, execution, and search. The SQL editor contains the following commands:

```
1 • CREATE DATABASE sri;
2 • USE sri;
3 • CREATE TABLE Customer (
4     Customer_ID INT PRIMARY KEY,
5     Customer_Name VARCHAR(100) NOT NULL,
6     Phone VARCHAR(15),
7     Email VARCHAR(100),
8     City VARCHAR(50)
9 );
10 • CREATE TABLE Account (
11     Account_No INT PRIMARY KEY,
12     Customer_ID INT,
13     Account_Type VARCHAR(20),
```

Below the editor is the 'Output' section, which is currently showing 'Action Output'. It contains a table with the following data:

#	Time	Action	Message
✓ 1	13:36:21	CREATE DATABASE sri	1 row(s) affected
✓ 2	13:36:21	USE sri	0 row(s) affected
✓ 3	13:36:46	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(10...	0 row(s) affected

```
CREATE TABLE Account (  
    Account_No INT PRIMARY KEY,  
    Customer_ID INT,  
    Account_Type VARCHAR(20),  
    Balance DECIMAL(12,2) DEFAULT 0,  
    Branch VARCHAR(50),  
    FOREIGN KEY (Customer_ID)  
    REFERENCES Customer(Customer_ID)  
);
```

Week 4 Triggers and Procedures

```

7      Email VARCHAR(100),
8      City VARCHAR(50)
9  );
10 CREATE TABLE Account (
11     Account_No INT PRIMARY KEY,
12     Customer_ID INT,
13     Account_Type VARCHAR(20),
14     Balance DECIMAL(12,2) DEFAULT 0,
15     Branch VARCHAR(50),
16     FOREIGN KEY (Customer_ID)
17     REFERENCES Customer(Customer_ID)
18 );
19 CREATE TABLE Bank_Transaction (

```

Output

Action Output

#	Time	Action	Message
1	13:36:21	CREATE DATABASE sri	1 row(s) affected
2	13:36:21	USE sri	0 row(s) affected
3	13:36:46	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(10...	0 row(s) affected
4	13:37:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Ty...	0 row(s) affected

CREATE TABLE Bank_Transaction (

Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,

Account_No INT,

Transaction_Type VARCHAR(20),

Amount DECIMAL(12,2),

Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

FOREIGN KEY (Account_No)

REFERENCES Account(Account_No)

);

```

18 ~ );
19 • CREATE TABLE Bank_Transaction (
20     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
21     Account_No INT,
22     Transaction_Type VARCHAR(20),
23     Amount DECIMAL(12,2),
24     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
25     FOREIGN KEY (Account_No)
26     REFERENCES Account(Account_No)
27 );
28 • CREATE TABLE Loan (

```

Output :

Action Output

#	Time	Action	Message
✓ 1	13:36:21	CREATE DATABASE sri	1 row(s) affected
✓ 2	13:36:21	USE sri	0 row(s) affected
✓ 3	13:36:46	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(10...	0 row(s) affected
✓ 4	13:37:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Ty...	0 row(s) affected
✓ 5	13:37:32	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected

CREATE TABLE Loan (

Loan_ID INT PRIMARY KEY,

Customer_ID INT,

Loan_Type VARCHAR(30),

Loan_Amount DECIMAL(12,2),

Interest_Rate DECIMAL(5,2),

FOREIGN KEY (Customer_ID)

REFERENCES Customer(Customer_ID)

);

```

27 );
28 CREATE TABLE Loan (
29     Loan_ID INT PRIMARY KEY,
30     Customer_ID INT,
31     Loan_Type VARCHAR(30),
32     Loan_Amount DECIMAL(12,2),
33     Interest_Rate DECIMAL(5,2),
34     FOREIGN KEY (Customer_ID)
35     REFERENCES Customer(Customer_ID)
36 );
37 INSERT INTO Customer

```

Output

Action Output

#	Time	Action	Message
1	13:36:21	CREATE DATABASE sri	1 row(s) affected
2	13:36:21	USE sri	0 row(s) affected
3	13:36:46	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(10...	0 row(s) affected
4	13:37:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Ty...	0 row(s) affected
5	13:37:32	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected
6	13:38:08	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARC...	0 row(s) affected

INSERT INTO Customer

(Customer_ID, Customer_Name, Phone, Email, City)

VALUES

(101, 'Ravi Kumar', '9876543210', 'ravi@gmail.com', 'Hyderabad'),

(102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),

(103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),

(104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),

(105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');

Limit to 1000 rows

```

37 • INSERT INTO Customer
38 (Customer_ID, Customer_Name, Phone, Email, City)
39 VALUES
40 (101, 'Ravi Kumar', '9876543210', 'ravi@gmail.com', 'Hyderabad'),
41 (102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),
42 (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
43 (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
44 (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
45 • INSERT INTO Account
46 (Account_No, Customer_ID, Account_Type, Balance, Branch)
47 VALUES
48 (10001, 101, 'Savings', 50000, 'Hyderabad'),
49 (10002, 102, 'Savings', 75000, 'Vijayawada'),

```

Output

Action Output

#	Time	Action	Message
2	13:36:21	USE sri	0 row(s) affected
3	13:36:46	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(1...	0 row(s) affected
4	13:37:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_...	0 row(s) affected
5	13:37:32	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,...	0 row(s) affected
6	13:38:08	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VAR...	0 row(s) affected
7	13:38:33	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Ravi ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

INSERT INTO Account

(Account_No, Customer_ID, Account_Type, Balance, Branch)

VALUES

(10001, 101, 'Savings', 50000, 'Hyderabad'),

(10002, 102, 'Savings', 75000, 'Vijayawada'),

(10003, 103, 'Current', 120000, 'Bangalore'),

(10004, 104, 'Savings', 45000, 'Chennai'),

(10005, 105, 'Current', 90000, 'Hyderabad');

```

44 (105, 'Kiran Kumar', 9876543214, 'Kiran@gmail.com', 'Hyderabad');
45 • INSERT INTO Account
46 (Account_No, Customer_ID, Account_Type, Balance, Branch)
47 VALUES
48 (10001, 101, 'Savings', 50000, 'Hyderabad'),
49 (10002, 102, 'Savings', 75000, 'Vijayawada'),
50 (10003, 103, 'Current', 120000, 'Bangalore'),
51 (10004, 104, 'Savings', 45000, 'Chennai'),
52 (10005, 105, 'Current', 90000, 'Hyderabad');
53 • INSERT INTO Bank_Transaction
54 (Account_No, Transaction_Type, Amount)
55 VALUES

```

Output

Action Output

#	Time	Action	Message
✓ 3	13:36:46	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(1...	0 row(s) affected
✓ 4	13:37:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_...	0 row(s) affected
✓ 5	13:37:32	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,...	0 row(s) affected
✓ 6	13:38:08	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VAR...	0 row(s) affected
✓ 7	13:38:33	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Ravi ...	5 row(s) affected Recd
✓ 8	13:38:57	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (1...	5 row(s) affected Recd

INSERT INTO Bank_Transaction

(Account_No, Transaction_Type, Amount)

VALUES

(10001, 'DEPOSIT', 10000),

(10002, 'DEPOSIT', 15000),

(10003, 'WITHDRAW', 20000),

(10004, 'DEPOSIT', 5000),

(10005, 'WITHDRAW', 10000);

```

54 (Account_No, Transaction_Type, Amount)
55 VALUES
56 (10001, 'DEPOSIT', 10000),
57 (10002, 'DEPOSIT', 15000),
58 (10003, 'WITHDRAW', 20000),
59 (10004, 'DEPOSIT', 5000),
60 (10005, 'WITHDRAW', 10000);
61 • INSERT INTO Loan
62 (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
63 VALUES
64 (501, 101, 'Home Loan', 5000000, 7.5),

```

Output



Action Output

	#	Time	Action
✓	4	13:37:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_...
✓	5	13:37:32	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,...
✓	6	13:38:08	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VAR...
✓	7	13:38:33	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Ravi ...
✓	8	13:38:57	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (1...
✓	9	13:39:33	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DE...

INSERT INTO Loan

(Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)

VALUES

(501, 101, 'Home Loan', 5000000, 7.5),

(502, 102, 'Education Loan', 1000000, 6.5),

(503, 103, 'Car Loan', 800000, 8.2),

(504, 104, 'Personal Loan', 500000, 10.5);


```

61 • INSERT INTO Loan
62 (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
63 VALUES
64 (501, 101, 'Home Loan', 5000000, 7.5),
65 (502, 102, 'Education Loan', 1000000, 6.5),
66 (503, 103, 'Car Loan', 800000, 8.2),
67 (504, 104, 'Personal Loan', 500000, 10.5);
68 • SELECT * FROM Customer;
69 • SELECT * FROM Account;
70 • SELECT * FROM Bank_Transaction;

```

Output

Action Output

#	Time	Action	Message
✓ 5	13:37:32	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,...	0 row(s) affected
✓ 6	13:38:08	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VAR...	0 row(s) affected
✓ 7	13:38:33	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Ravi ...	5 row(s) affected
✓ 8	13:38:57	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (1...	5 row(s) affected
✓ 9	13:39:33	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DE...	5 row(s) affected
✓ 10	13:39:58	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (...	4 row(s) affected

SELECT * FROM Customer;

```

68 • SELECT * FROM Customer;

```

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
NULL	NULL	NULL	NULL	NULL

SELECT * FROM Account;

```

69 • SELECT * FROM Account;
70 • SELECT * FROM Bank_Transaction;
71 • SELECT * FROM Loan;

```

Result Grid					
Filter Rows:					
	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
▲	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Bank_Transaction;

```

70 • SELECT * FROM Bank_Transaction;
71 • SELECT * FROM Loan;

```

Result Grid					
Filter Rows:					
	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
▶	1	10001	DEPOSIT	10000.00	2026-08-14 13:39:33
	2	10002	DEPOSIT	15000.00	2026-08-14 13:39:33
	3	10003	WITHDRAW	20000.00	2026-08-14 13:39:33
	4	10004	DEPOSIT	5000.00	2026-08-14 13:39:33
	5	10005	WITHDRAW	10000.00	2026-08-14 13:39:33
▲	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Loan;

71 •

```
SELECT * FROM Loan;
```

Result Grid					
			 Filter Rows:	Edit:   	
	Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_f
	501	101	Home Loan	5000000.00	7.50
	502	102	Education Loan	1000000.00	6.50
	503	103	Car Loan	800000.00	8.20
	504	104	Personal Loan	500000.00	10.50
	NULL	NULL	NULL	NULL	NULL

```
DELIMITER //
```

```
CREATE PROCEDURE GetAllCustomers()
```

```
BEGIN
```

```
    SELECT * FROM Customer;
```

```
END //
```

```
DELIMITER ;
```

```
CALL GetAllCustomers();
```

```
DELIMITER //
```

```
CREATE PROCEDURE GetAccountDetails(
```

```
    IN p_Account_No INT
```

```
)
```

```
BEGIN
```

```
    SELECT *
```

```
    FROM Account
```

```
    WHERE Account_No = p_Account_No;
```

```
END //
```

```
DELIMITER ;
```

```
CALL GetAccountDetails(10001);
```

```

83 BEGIN
84     SELECT *
85     FROM Account
86     WHERE Account_No = p_Account_No;
87 END //
--

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	Customer_ID	Customer_Name	Phone	Email	City
▶	101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
	102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
	103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
	104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
	105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

DELIMITER //

CREATE PROCEDURE GetCustomerAccounts(

IN p_Customer_ID INT

)

BEGIN

SELECT

C.Customer_ID,

C.Customer_Name,

A.Account_No,

A.Account_Type,

A.Balance,

A.Branch

FROM Customer C

JOIN Account A

ON C.Customer_ID = A.Customer_ID

WHERE C.Customer_ID = p_Customer_ID;

END //

DELIMITER ;

CALL GetCustomerAccounts(101);

Execute the selected portion of the script or everything, if there is no selection

```
105 WHERE c.Customer_ID = p_Customer_ID;  
106 END //  
107 DELIMITER ;  
108 • CALL GetCustomerAccounts(101);  
...
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
▶	101	Ravi Kumar	10001	Savings	50000.00	Hyderabad

DELIMITER //

CREATE PROCEDURE DepositMoney(

IN p_Account_No INT,

IN p_Amount DECIMAL(12,2)

)

BEGIN

UPDATE Account

SET Balance = Balance + p_Amount

WHERE Account_No = p_Account_No;

END //

DELIMITER ;

CALL DepositMoney(10001, 5000);

```

114 BEGIN
115     UPDATE Account
116     SET Balance = Balance + p_Amount
117     WHERE Account_No = p_Account_No;
118 END //
119 DELIMITER ;
120 CALL DepositMoney(10001, 5000);
121 DELIMITER //
122 CREATE PROCEDURE WithdrawMoney(
123     IN p_Account_No INT,
124     IN p_Amount DECIMAL(12,2)

```

Output :

#	Time	Action
✓ 18	13:43:50	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT *
✓ 19	13:47:30	CALL GetAccountDetails(10001)
✓ 20	13:47:30	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELECT.
✓ 21	13:47:30	CALL GetCustomerAccounts(101)
✓ 22	13:48:10	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(...
✓ 23	13:48:10	CALL DepositMoney(10001, 5000)

DELIMITER //

CREATE PROCEDURE WithdrawMoney(

IN p_Account_No INT,

IN p_Amount DECIMAL(12,2)

)

BEGIN

UPDATE Account

SET Balance = Balance - p_Amount

WHERE Account_No = p_Account_No;

END //

DELIMITER ;

CALL WithdrawMoney(10001, 3000);

Execute the selected portion of the script or everything, if there is no selection

```

121 DELIMITER //
122 CREATE PROCEDURE WithdrawMoney(
123     IN p_Account_No INT,
124     IN p_Amount DECIMAL(12,2)
125 )
126 BEGIN
127     UPDATE Account
128     SET Balance = Balance - p_Amount
129     WHERE Account_No = p_Account_No;
130 END //
131 DELIMITER ;
132 CALL WithdrawMoney(10001, 3000);
133 DELIMITER //

```

Output

Action Output

#	Time	Action	Message
✓ 20	13:47:30	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELECT...	0 rows
✓ 21	13:47:30	CALL GetCustomerAccounts(101)	1 row
✓ 22	13:48:10	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(...	0 rows
✓ 23	13:48:10	CALL DepositMoney(10001, 5000)	1 row
✓ 24	13:48:50	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMA...	0 rows
✓ 25	13:48:50	CALL WithdrawMoney(10001, 3000)	1 row

DELIMITER //

CREATE TRIGGER CheckBalance

BEFORE UPDATE ON Account

FOR EACH ROW

BEGIN

IF NEW.Balance < 0 THEN

SIGNAL SQLSTATE '45000'

SET MESSAGE_TEXT =

'Transaction failed: Insufficient balance';

END IF;

END //

DELIMITER ;

UPDATE Account

SET Balance = Balance - 60000

WHERE Account_No = 10001;

Expected Result:

Transaction failed: Insufficient balance

✖ 43 13:57:53 UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Transaction failed: Insufficient balance
---	--

DELIMITER //

CREATE TRIGGER CheckTransactionAmount

BEFORE INSERT ON Bank_Transaction

FOR EACH ROW

BEGIN

IF NEW.Amount <= 0 THEN

SIGNAL SQLSTATE '45000'

SET MESSAGE_TEXT =

'Transaction amount must be greater than zero';

END IF;

END //

33 13:57:40 INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001,...	Error Code: 1644. Transaction amount must be greater than zero
---	--

DELIMITER ;

INSERT INTO Bank_Transaction

(Account_No, Transaction_Type, Amount)

VALUES

(10001, 'DEPOSIT', -5000);

CREATE TABLE Transaction_Audit (

Audit_ID INT PRIMARY KEY AUTO_INCREMENT,

Transaction_ID INT,

Account_No INT,

Transaction_Type VARCHAR(20),

Amount DECIMAL(12,2),

Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP

);

✖	33	13:57:40	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001,...	Error Code: 1644. Ir
✔	34	14:00:50	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, ...	0 row(s) affected

DELIMITER //

CREATE TRIGGER TransactionAudit

AFTER INSERT ON Bank_Transaction

FOR EACH ROW

BEGIN

INSERT INTO Transaction_Audit

(

Transaction_ID,

Account_No,

Transaction_Type,

Amount

)

VALUES

(

NEW.Transaction_ID,

NEW.Account_No,

NEW.Transaction_Type,

NEW.Amount

);

END //

DELIMITER ;

INSERT INTO Bank_Transaction

(Account_No, Transaction_Type, Amount)

VALUES

(10001, 'DEPOSIT', 2500);

SELECT * FROM Transaction_Audit;

✔	36	14:02:07	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001,...	1 row(s) affected
✔	37	14:02:07	SELECT * FROM Transaction_Audit LIMIT 0, 1000	1 row(s) returned

Result Grid						
		Filter Rows:		Edit:		Export/Import:
	Audit_ID	Transaction_ID	Account_No	Transaction_Type	Amount	Audit_Date
▶	1	6	10001	DEPOSIT	2500.00	2026-08-14 14:02:07
*	NULL	NULL	NULL	NULL	NULL	NULL

DELIMITER //

CREATE TRIGGER UpdateBalanceAfterTransaction

AFTER INSERT ON Bank_Transaction

FOR EACH ROW

BEGIN

IF NEW.Transaction_Type = 'DEPOSIT' THEN

UPDATE Account

SET Balance = Balance + NEW.Amount

WHERE Account_No = NEW.Account_No;

ELSEIF NEW.Transaction_Type = 'WITHDRAW' THEN

UPDATE Account

SET Balance = Balance - NEW.Amount

WHERE Account_No = NEW.Account_No;

END IF;

END //

DELIMITER ;

INSERT INTO Bank_Transaction

(Account_No, Transaction_Type, Amount)

VALUES

(10001, 'DEPOSIT', 5000);

SELECT *

FROM Account

WHERE Account_No = 10001;

220	FROM Account
221	WHERE Account_No = 10001;
222	DELIMITER //

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap
Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	54000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

DELIMITER //

CREATE TRIGGER PreventInsufficientWithdrawal

BEFORE INSERT ON Bank_Transaction

FOR EACH ROW

BEGIN

DECLARE CurrentBalance DECIMAL(12,2);

SELECT Balance

INTO CurrentBalance

FROM Account

WHERE Account_No = NEW.Account_No;

IF NEW.Transaction_Type = 'WITHDRAW'

AND NEW.Amount > CurrentBalance THEN

SIGNAL SQLSTATE '45000'

SET MESSAGE_TEXT =

'Withdrawal failed: Insufficient balance';

END IF;

END //

DELIMITER ;

INSERT INTO Bank_Transaction

(Account_No, Transaction_Type, Amount)

VALUES

(10001, 'WITHDRAW', 1000000);

i2 INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001,... Error Code: 1644. Withdrawal failed: Insufficient balance

DELIMITER //

```

CREATE PROCEDURE TransferMoney(
    IN SenderAccount INT,
    IN ReceiverAccount INT,
    IN TransferAmount DECIMAL(12,2)
)
BEGIN
    DECLARE SenderBalance DECIMAL(12,2);
    SELECT Balance
    INTO SenderBalance
    FROM Account
    WHERE Account_No = SenderAccount;
    IF SenderBalance < TransferAmount THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT =
        'Transfer failed: Insufficient balance';
    ELSE
        UPDATE Account
        SET Balance = Balance - TransferAmount
        WHERE Account_No = SenderAccount;
        UPDATE Account
        SET Balance = Balance + TransferAmount
        WHERE Account_No = ReceiverAccount;
    END IF;
END //
DELIMITER ;
CALL TransferMoney(10001, 10002, 5000);
SELECT *
FROM Account
WHERE Account_No IN (10001,10002);

```

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	49000.00	Hyderabad
	10002	102	Savings	80000.00	Vijayawada
•	NULL	NULL	NULL	NULL	NULL

DELIMITER //

CREATE PROCEDURE GetCustomerLoans(

IN p_Customer_ID INT

)

BEGIN

SELECT

C.Customer_Name,

L.Loan_ID,

L.Loan_Type,

L.Loan_Amount,

L.Interest_Rate

FROM Customer C

JOIN Loan L

ON C.Customer_ID = L.Customer_ID

WHERE C.Customer_ID = p_Customer_ID;

END //

DELIMITER ;

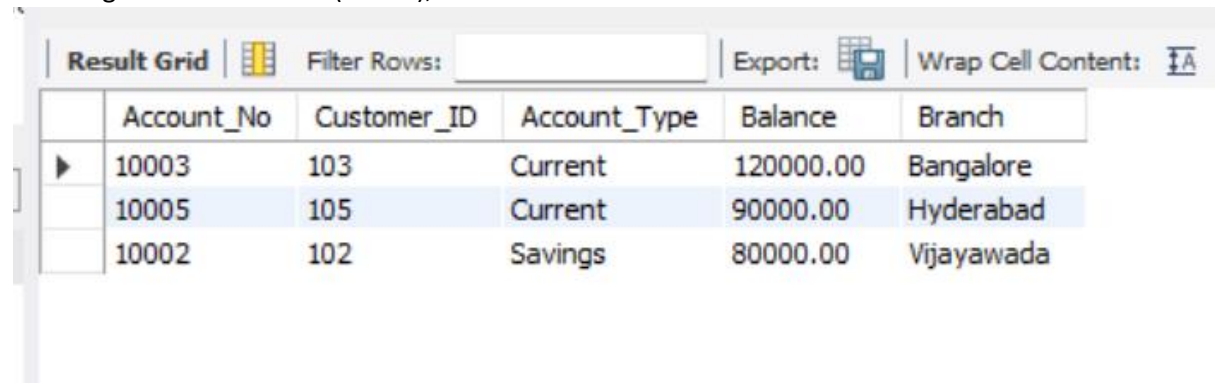
CALL GetCustomerLoans(101);

Result Grid					
Filter Rows:					
Export:					
Wrap Cell Content:					
	Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
▶	Ravi Kumar	501	Home Loan	5000000.00	7.50

DELIMITER //

CREATE PROCEDURE HighBalanceAccounts(

```
IN MinimumBalance DECIMAL(12,2)
)
BEGIN
SELECT *
FROM Account
WHERE Balance >= MinimumBalance
ORDER BY Balance DESC;
END //
DELIMITER ;
CALL HighBalanceAccounts(50000);
```



The screenshot shows a database result grid with the following columns: Account_No, Customer_ID, Account_Type, Balance, and Branch. The grid contains three rows of data. The first row has Account_No 10003, Customer_ID 103, Account_Type Current, Balance 120000.00, and Branch Bangalore. The second row has Account_No 10005, Customer_ID 105, Account_Type Current, Balance 90000.00, and Branch Hyderabad. The third row has Account_No 10002, Customer_ID 102, Account_Type Savings, Balance 80000.00, and Branch Vijayawada. The grid also includes a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox.

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10003	103	Current	120000.00	Bangalore
	10005	105	Current	90000.00	Hyderabad
	10002	102	Savings	80000.00	Vijayawada